

SECTION 17.1

6. $\oint_C \mathbf{F} \cdot d\mathbf{r}$, where $\mathbf{F}(x, y) = (x + y, x^2 - y)$ and C is the boundary of the region enclosed by $y = x^2$ and $y = \sqrt{x}$ for $0 \leq x \leq 1$

$$I = \int_C \mathbf{F} \cdot d\mathbf{s} = \iint_D \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} \right) \cdot dA$$

$$P = x + y \quad Q = x^2 - y$$

$$\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} = 2x - 1$$

So,

$$I = \iint_D (2x - 1) \cdot dA = \int_0^1 \int_{x^2}^{\sqrt{x}} (2x - 1) \cdot dy \cdot dx$$

$$= \int_0^1 (2x - 1)y \Big|_{y=x^2}^{\sqrt{x}} \cdot dx$$

$$= \int_0^1 (2x - 1)(\sqrt{x} \cdot x^2) \cdot dx = \int_0^1 (2x^{3/2} - 2x^3 - x^{1/2} + x^2) \cdot dx$$

$$= \frac{4}{5} x^{5/2} - \frac{1}{2} x^4 - \frac{2}{3} x^{3/2} + \frac{x^3}{3} \Big|_0^1 = \frac{4}{5} - \frac{1}{2} - \frac{2}{3} + \frac{1}{3}$$

$$= -1/30$$

8. $\oint_C (\ln x + y) dx - x^2 dy$, where C is the rectangle with vertices (1, 1), (3, 1), (1, 4), and (3, 4)

$$I = \int_C \mathbf{F} \cdot d\mathbf{s} = \iint_D \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} \right) \cdot dA$$

$$P = \ln(x + y) \quad Q = -x^2$$

$$\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} = -2x - 1$$

$$I = \iint_D (-2x - 1) \cdot dA = \int_1^3 \int_1^4 (-2x - 1) \cdot dy \cdot dx$$

$$= \int_1^3 (-2x - 1)y \Big|_1^4 \cdot dx = \int_1^3 -3(2x + 1) \cdot dx$$

$$= -3(12 - 2) = -30$$

11. Let $\mathbf{F}(x, y) = (2xe^y, x + x^2e^y)$ and let C be the quarter-circle path from A to B in Figure 18. Evaluate $I = \oint_C \mathbf{F} \cdot d\mathbf{r}$ as follows:

(a) Find a function $f(x, y)$ such that $\mathbf{F} = \mathbf{G} + \nabla f$, where $\mathbf{G} = \langle 0, x \rangle$.

(b) Show that the line integrals of $\mathbf{G}$ along the segments $\overline{OA}$ and $\overline{OB}$ are zero.

(c) Evaluate I . Hint: Use Green's Theorem to show that

$$I = f(B) - f(A) + 4\pi$$

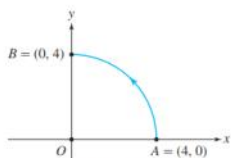


FIGURE 18

a) $\mathbf{F} - \mathbf{G} = \langle 2xe^y, x + x^2e^y \rangle - \langle 0, x \rangle = \langle 2xe^y, x^2e^y \rangle$

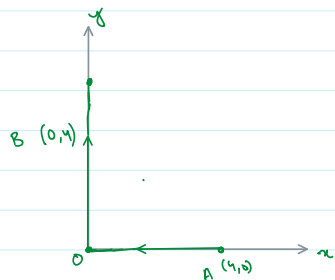
$$V(x, y) = x^2e^y$$

b) $\overline{AO} : \langle t, 0 \rangle, 0 \leq t \leq 4$ $\overline{OB} : \langle 0, t \rangle : 0 \leq t \leq 4$

$$\mathbf{G} = \langle 0, x \rangle.$$

$$\int_{\overline{OA}} \mathbf{G} \cdot d\mathbf{s} = \int_0^4 \langle 0, t \rangle \cdot \langle 1, 0 \rangle \cdot dt = \int_0^4 0 \cdot dt = 0$$

$$\int_{\overline{OB}} \mathbf{G} \cdot d\mathbf{s} = \int_0^4 \langle 0, 0 \rangle \cdot \langle 0, 1 \rangle \cdot dt = \int_0^4 0 \cdot dt = 0$$



c) $\mathbf{F} - \mathbf{G} = \nabla V,$

$$\int_C (\mathbf{F} - \mathbf{G}) \cdot d\mathbf{s} = V(B) - V(A) = \int_C \mathbf{F} \cdot d\mathbf{s} - \int_C \mathbf{G} \cdot d\mathbf{s}$$

$$= 1 - \int_C \mathbf{G} \cdot d\mathbf{s}$$

$$1 = V(B) - V(A) + \int_C \mathbf{G} \cdot d\mathbf{s}$$

$$\int_C \mathbf{G} \cdot d\mathbf{s} = \int_{\overline{BO} + \overline{OB} + \overline{OC}} \mathbf{G} \cdot d\mathbf{s} = \int_{\overline{BO}} \mathbf{G} \cdot d\mathbf{s} - \int_{\overline{OB}} \mathbf{G} \cdot d\mathbf{s}$$

Now,

$$\int_C \mathbf{G} \cdot d\mathbf{s} = \int \mathbf{G} \cdot d\mathbf{s}$$

Now,

$$\int_C G \cdot ds = \int_{\sqrt{36+9x^2+4}} G \cdot ds$$

Using Green's theorem:

$$G = (0, x), \quad P = 0 \quad Q = x$$

$$\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} = 1 - 0 = 1.$$

$$\int_{\sqrt{36+9x^2+4}} G \cdot ds = \iint_D 1 \cdot dA = \text{Area}(D) = \boxed{4\pi} \text{ Am.}$$

So,

$$I = V(B) - V(A) + 4\pi$$

$$I = V(0, 4) - V(4, 0) + 4\pi = 0 - 4^2 e^0 + 4\pi = \boxed{4\pi - 16} \text{ Am.}$$

16. The region between the x-axis and the cycloid parametrized by $\mathbf{r}(t) = (t - \sin t, 1 - \cos t)$ for $0 \leq t \leq 2\pi$ (Figure 21)



FIGURE 21 Cycloid.

$$A = \frac{1}{2} \int_C x \, dy - y \, dx + \frac{1}{2} \int_C x \, dy - y \, dx$$

$$x \, dy - y \, dx = t \cdot 0 \, dt - 0 \cdot dt = 0$$

$$\int_{\text{on}} x \, dy - y \, dx = 0$$

$$x = t - \sin t \quad y = 1 - \cos t, \text{ therefore}$$

$$\begin{aligned} x \, dy - y \, dx &= (t - \sin t) \sin t \cdot dt - (1 - \cos t)(1 - \cos t) \, dt \\ &= (t \sin t - \sin^2 t - 1 + 2 \cos t - \cos^2 t) \cdot dt = (t \sin t + 2 \cos t - 2) \, dt \end{aligned}$$

$$\int_C x \, dy - y \, dx = \int_{2\pi}^0 (t \sin t + 2 \cos t - 2) \, dt = \int_0^{2\pi} (2 - 2 \cos t - t \sin t) \, dt$$

$$= 2t - 2 \sin t + t \cos t - \sin t \Big|_0^{2\pi} = 2t - 3 \sin t + t \cos t \Big|_0^{2\pi}$$

$$\boxed{= 6\pi}$$

$$A = \frac{1}{2} \cdot 0 + \frac{1}{2} \cdot 6\pi = 3\pi$$

19. Let $x^3 + y^3 = 3xy$ be the folium of Descartes (Figure 22).

(a) Show that the folium has a parametrization in terms of $t = y/x$ given by

$$x = \frac{3t}{1+t^3}, \quad y = \frac{3t^2}{1+t^3} \quad (-\infty < t < \infty) \quad (t \neq -1)$$

(b) Show that

$$x \, dy - y \, dx = \frac{9t^2}{(1+t^3)^2} \, dt$$

Hint: By the Quotient Rule,

$$x^2 d\left(\frac{y}{x}\right) = x \, dy - y \, dx$$

(c) Find the area of the loop of the folium. Hint: The limits of integration are 0 and ∞ .

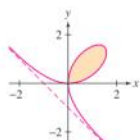


FIGURE 22 Folium of Descartes.

a)

$$x = \frac{3t}{1+t^3}, \quad y = \frac{3t^2}{1+t^3} \quad x^3 + y^3 - 3xy = 0$$

$$\begin{aligned} x^3 + y^3 - 3xy &= \left(\frac{3t}{1+t^3}\right)^3 + \left(\frac{3t^2}{1+t^3}\right)^3 - 3 \cdot \frac{3t}{1+t^3} \cdot \frac{3t^2}{1+t^3} \\ &= \frac{27t^3 + 27t^6}{(1+t^3)^3} - \frac{27t^3(1+t^3)}{(1+t^3)^3} = \frac{27t^3(1+t^3 - (1+t^3))}{(1+t^3)^3} = 0 \end{aligned}$$

$$x = \frac{3t}{1+t^3} \quad y = \frac{3t^2}{1+t^3}$$

$$t = \frac{y}{x}$$

b)

$$I = \frac{x \frac{dy}{dt} - y \frac{dx}{dt}}{x^2}$$

$$x \frac{dy}{dt} - y \frac{dx}{dt} = x^2 = \left(\frac{3t}{1+t^3}\right)^2$$

$$x \, dy - y \, dx = \frac{9t^2}{(1+t^3)^2} \cdot dt$$

c)

$$A = \frac{1}{2} \int_c x \, dy - y \, dx = \frac{1}{2} \int_0^\infty \frac{9t^2}{(1+t^3)^2} \cdot dt$$

$$A = \frac{1}{2} \lim_{R \rightarrow \infty} \int_0^R \frac{9t^2}{(1+t^3)^2} \cdot dt = \frac{1}{2} \lim_{R \rightarrow \infty} \int_1^{1+R^3} \frac{3}{u^2} \, du$$

$$= \frac{3}{2} \lim_{R \rightarrow \infty} \left. -\frac{1}{u} \right|_{u=1}^{1+R^3} = \frac{3}{2} (1-0) = \boxed{\frac{3}{2}} \text{ An.}$$

23. Let C_R be the circle of radius R centered at the origin. Use the general form of Green's Theorem to determine $\oint_{C_2} \mathbf{F} \cdot d\mathbf{r}$, where $\mathbf{F}$ is a vector field such that $\oint_{C_1} \mathbf{F} \cdot d\mathbf{r} = 9$ and $\frac{\partial F_2}{\partial x} - \frac{\partial F_1}{\partial y} = x^2 + y^2$ for (x, y) in the annulus $1 \leq x^2 + y^2 \leq 4$.

$$\int_{C_2} \mathbf{F} \cdot d\mathbf{s} - \int_{C_1} \mathbf{F} \cdot d\mathbf{s} = \iint_D \left(\frac{\partial F_2}{\partial x} - \frac{\partial F_1}{\partial y} \right) \cdot dx \cdot dy$$

$$\int_{C_2} \mathbf{F} \cdot d\mathbf{s} - 9 = \iint_D (x^2 + y^2) \cdot dx \cdot dy$$

$$\int_{C_2} \mathbf{F} \cdot d\mathbf{s} = 9 + \iint_D (x^2 + y^2) \cdot dx \cdot dy$$

$$\int_{C_2} \mathbf{F} \cdot d\mathbf{s} = 9 + \int_0^{2\pi} \int_1^2 r^2 \cdot r \cdot dr \cdot d\theta = 9 + 2\pi \int_1^2 r^3 \cdot dr = 9 + 2\pi \left[\frac{r^4}{4} \right]_1^2$$

$$= 9 + \frac{15\pi}{2} \quad \text{Ans}$$

25. Referring to Figure 25, suppose that

$$\oint_{C_2} \mathbf{F} \cdot d\mathbf{r} = 3\pi, \quad \oint_{C_3} \mathbf{F} \cdot d\mathbf{r} = 4\pi$$

Use Green's Theorem to determine the circulation of $\mathbf{F}$ around C_1 , assuming that $\frac{\partial F_2}{\partial x} - \frac{\partial F_1}{\partial y} = 9$ on the shaded region.

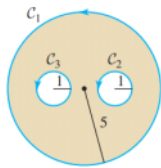


FIGURE 25

$$\int_{C_1} \mathbf{F} \cdot d\mathbf{s} - \int_{C_2} \mathbf{F} \cdot d\mathbf{s} - \int_{C_3} \mathbf{F} \cdot d\mathbf{s} = \iint_D \text{curl}(\mathbf{F}) \cdot d\mathbf{A}$$

$$\int_{C_1} \mathbf{F} \cdot d\mathbf{s} - 3\pi - 4\pi = \iint_D 9 \cdot dA = 9 \iint_D 1 \cdot dA = 9 \cdot \text{Area}(D)$$

$$\text{Area}(D) = \pi \cdot 5^2 - \pi \cdot 1^2 - \pi \cdot 1^2 = \boxed{23\pi}$$

$$\int_{C_1} \mathbf{F} \cdot d\mathbf{s} = 7\pi = 9 \cdot 23\pi = \boxed{207\pi}$$

26. Let $\mathbf{F}$ be the vortex vector field

$$\mathbf{F}(x, y) = \left(\frac{-y}{x^2 + y^2}, \frac{x}{x^2 + y^2} \right)$$

In Section 16.3, we verified that $\int_{C_R} \mathbf{F} \cdot d\mathbf{r} = 2\pi$, where C_R is the circle of radius R centered at the origin. Prove that $\oint_C \mathbf{F} \cdot d\mathbf{r} = 2\pi$ for any simple closed curve C whose interior contains the origin (Figure 26). *Hint:* Apply the general form of Green's Theorem to the domain between C and C_R , where R is so small that C_R is contained in C .

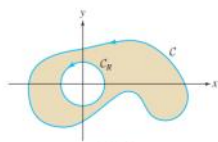


FIGURE 26

In Exercises 27–30, we refer to the integrand that occurs in Green's Theorem and that appears as

$$\text{curl}_z(\mathbf{F}) = \frac{\partial F_2}{\partial x} - \frac{\partial F_1}{\partial y}$$

$$\int_C \mathbf{F} \cdot d\mathbf{s} + \int_{C_R} \mathbf{F} \cdot d\mathbf{s} = \iint_D \text{curl}_z(\mathbf{F}) \cdot d\mathbf{A}$$

$$\int_C \mathbf{F} \cdot d\mathbf{s} = 2\pi = \iint_D 0 \cdot d\mathbf{A} = 0$$

$$\boxed{\int_C \mathbf{F} \cdot d\mathbf{s} = 2\pi}$$

SECTION 17.2

4. $\mathbf{F} = \langle y, x, x^2 + y^2 \rangle$, the upper hemisphere
 $x^2 + y^2 + z^2 = 1, z \geq 0$

$$\gamma(t) = \langle \cos t, \sin t, 0 \rangle, 0 \leq t \leq 2\pi$$

$$\mathbf{F}(\gamma(t)) \cdot \gamma'(t) = \langle \sin t, \cos t, 2 \rangle \cdot \langle -\sin t, \cos t, 0 \rangle = -\sin^2 t + 2\cos t = \cos 2t$$

$$\int_C \mathbf{F} \cdot d\mathbf{s} = \int_0^{2\pi} (\cos^2 t - \sin^2 t) dt = \int_0^{2\pi} \cos 2t \cdot dt = \left. \frac{1}{2} \sin 2t \right|_0^{2\pi} = 0$$

$$\text{curl}(\mathbf{F}) = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ y & x & x^2 + y^2 \end{vmatrix} = \langle 2xy, -2x, 0 \rangle$$

$$\mathbf{r}(\phi, \theta) = \langle \cos \theta \sin \phi, \sin \theta \sin \phi, \cos \phi \rangle, \quad 0 \leq \theta \leq 2\pi \\ 0 \leq \phi \leq \pi/2$$

$$\mathbf{r}_\theta = \langle -\sin \theta \sin \phi, \cos \theta \sin \phi, 0 \rangle$$

$$\mathbf{r}_\phi = \langle \cos \theta \cos \phi, \sin \theta \cos \phi, -\sin \phi \rangle$$

$$\mathbf{r}_\theta \times \mathbf{r}_\phi = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ -\sin \theta \sin \phi & \cos \theta \sin \phi & 0 \\ \cos \theta \cos \phi & \sin \theta \cos \phi & -\sin \phi \end{vmatrix}$$

$$= \left\langle -\cos Q \sin \phi, -\sin Q \sin \phi, -\frac{\sin^2 Q \sin 2\phi}{2} - \frac{\cos^2 Q \sin 2\phi}{2} \right\rangle$$

$$n = -G_Q \times G_\phi$$

$$n = \sin \phi \langle \cos Q \sin \phi, \sin Q \sin \phi, \cos \phi \rangle$$

$$\text{curl}(F)(G(\phi, Q)) = \langle 2 \sin Q \sin \phi, -2 \cos Q \sin \phi, 0 \rangle$$

$$\text{curl}(F) \cdot n = \sin Q (2 \sin Q \cos Q \sin^2 \phi - 2 \sin Q \cos Q \sin^2 \phi + 0) = 0$$

So,

$$\iint_D 0 \cdot dA = \boxed{0} \text{ Am.}$$

6. $F = (x + y, z^2 - 4, x\sqrt{y^2 + 1})$, surface of the wedge-shaped box in Figure 15 (bottom included, top excluded) with outward-pointing normal

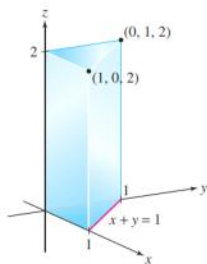


FIGURE 15

$$F = \langle x+y, 0, x\sqrt{y^2+1} \rangle$$

parametrize: $0 \leq t \leq 1$:

$$\langle 1-t, 0, 0 \rangle, \langle 0, t, 0 \rangle, \langle t, 1-t, 0 \rangle$$

This ds on three sides:

$$\langle -1, 0, 0 \rangle dt, \langle 0, 1, 0 \rangle dt, \langle 1, -1, 0 \rangle dt$$

$$F \cdot ds = \langle 1-t, 0, 0 \rangle \cdot \langle -1, 0, 0 \rangle dt = (t-1) \cdot dt$$

$$F \cdot ds = \langle t, 0, 0 \rangle \cdot \langle 0, 1, 0 \rangle dt = 0$$

$$F \cdot ds = \langle 1, 0, 0 \rangle \cdot \langle 1, -1, 0 \rangle dt = dt$$

$$\int_0^1 (t-1) dt + 0 + \int_0^1 dt = -\frac{1}{2} + 0 + 1 = \boxed{\frac{1}{2}}$$

12. $F = \langle yz, xy, xz \rangle$, C is the square with vertices $(0, 0, 2)$, $(1, 0, 2)$, $(1, 1, 2)$, and $(0, 1, 2)$, oriented counterclockwise as viewed from above.

$$\bullet C_1: (0, 0, 2) \rightarrow (1, 0, 2)$$

- $C_2: (1, 0, z) \rightarrow (1, 1, 2)$

- $C_3: (1, 1, 2) \rightarrow (0, 1, 2)$

- $C_4: (0, 1, 2) \rightarrow (0, 0, 2)$

- Edge $C_1: r = (t, 0, 2), t \in [0, 1]$

$$dr = (1, 0, 0) dt$$

$$F = (0.2, t.0, t.2) = (0, 0, 2t)$$

$$F \cdot dr = 0 \cdot dt$$

$$\int_{C_1} = 0$$

- Edge $C_2: r = (1, t, 2), t \in [0, 1]$

$$dr = (0, 1, 0) dt$$

$$F = (t.2, 1.t, 1.2) = (2t, t, 2)$$

$$F \cdot dr = t \cdot dt$$

$$\int_{C_2} dA = \int_0^1 t \cdot dt = 1/2$$

- Edge $C_3: r = (1-t, 1, 2), t \in [0, 1]$

$$dr = (-1, 0, 0) dt$$

$$F = (1.2, (1-t), (1-t) \cdot 2) = (2, 1-t, 2-2t)$$

- edge $C_4: r = (0, 1-t, 2), t \in [0, 1]$

$$dr = (0, -1, 0) dt$$

$$F = ((1-t) \cdot 2, 0, 0) = (2-2t, 0, 0)$$

$$F \cdot dr = 0 \cdot dt = 0$$

- Total = $\oint_{C_F} dr = 0 + 1/2 - 2 + 0 = -3/2$

- Surface integral $\iint_S (\nabla \times F) \cdot n \, ds$

$$\text{curl of } F: \nabla \times F = \begin{vmatrix} i & j & k \\ \partial/\partial x & \partial/\partial y & \partial/\partial z \\ yz & xy & xz \end{vmatrix} = (0, y-z, y-z)$$

Now:-

$$(\nabla \times \mathbf{F}) \cdot \mathbf{n} = (0, y-z, y-z) \cdot (0, 0, 1) = y-z$$

$$\text{Surface Integral} = \iint_S (y-z) \cdot dA \quad \begin{matrix} 0 \leq x \leq 1 \\ 0 \leq y \leq 1 \end{matrix}$$

$$= \int_0^1 \int_0^1 (y-z) dx dy = \int_0^1 (y-z) dy = \left[\frac{y^2}{2} - zy \right]_0^1 = \frac{-3}{2}$$

• Hence Stokes' theorem is verified.

15. Let S be the surface of the cylinder (not including the top and bottom) of radius 2 for $1 \leq z \leq 6$, oriented with outward-pointing normal (Figure 16).

(a) Indicate with an arrow the orientation of ∂S (the top and bottom circles).

(b) Verify Stokes' Theorem for S and $\mathbf{F} = \langle yz^2, 0, 0 \rangle$.

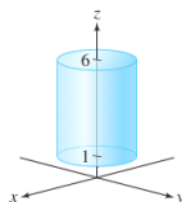


FIGURE 16

$$C_1: \gamma_1(t) = (2 \cos t, 2 \sin t, 6), \quad 0 \leq t \leq 2\pi$$

$$C_2: \gamma_2(t) = (2 \cos t, 2 \sin t, 1), \quad 0 \leq t \leq 2\pi$$

$$\mathbf{F}(\gamma_1(t)) = \langle yz^2, 0, 0 \rangle = \langle 72 \sin t, 0, 0 \rangle$$

$$\gamma_1'(t) = \langle -2 \sin t, 2 \cos t, 0 \rangle$$

$$\begin{aligned} \mathbf{F}(\gamma_1(t)) \cdot \gamma_1'(t) &= \langle 72 \sin t, 0, 0 \rangle \cdot \langle -2 \sin t, 2 \cos t, 0 \rangle \\ &= -144 \sin^2 t \end{aligned}$$

$$\mathbf{F}(\gamma_2(t)) = \langle yz^2, 0, 0 \rangle = \langle 2 \sin t, 0, 0 \rangle$$

$$\gamma_2'(t) = \langle -2 \sin t, 2 \cos t, 0 \rangle$$

$$\begin{aligned} \mathbf{F}(\gamma_2(t)) \cdot \gamma_2'(t) &= \langle 2 \sin t, 0, 0 \rangle \cdot \langle -2 \sin t, 2 \cos t, 0 \rangle \\ &= -4 \sin^2 t \end{aligned}$$

$$\int_C \mathbf{F} \cdot d\mathbf{s} = \int_{C_1} \mathbf{F} \cdot d\mathbf{s} + \int_{C_2} \mathbf{F} \cdot d\mathbf{s} = \int_0^{2\pi} (-144 \sin^2 t) dt + \int_0^{2\pi} (-4 \sin^2 t) dt$$

$$= \int_0^{2\pi} 140 \sin^2 t \cdot dt = 140 \int_0^{2\pi} \frac{1 - \cos 2t}{2} \cdot dt = \boxed{140\pi} \text{ Ans.}$$

• $\text{curl}(\mathbf{F})$:-

$$\nabla \times \mathbf{F} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ x^2 y & y^2 z & -z^2 \end{vmatrix} = \langle 0, 2yz, -2z^2 \rangle$$

$$\nabla \times \mathbf{F} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ \partial/\partial x & \partial/\partial y & \partial/\partial z \\ yz^2 & 0 & 0 \end{vmatrix} = \langle 0, 2yz, -z^2 \rangle$$

• Parametrize S by:-

$$\phi(\theta, z) = (2\cos\theta, 2\sin\theta, z) \quad 0 \leq \theta \leq 2\pi$$

$$1 \leq z \leq 6$$

$$\mathbf{n} = \langle 2\cos\theta, 2\sin\theta, 0 \rangle$$

$$\text{curl}(\mathbf{F}) \cdot (\phi(\theta, z)) \cdot \mathbf{n} = \langle 0, 4z\sin\theta, -z^2 \rangle \cdot \langle 2\cos\theta, 2\sin\theta, 0 \rangle = 8z\sin\theta$$

$$\iint_S \text{curl}(\mathbf{F}) \cdot d\mathbf{S} = \int_1^6 \int_0^{2\pi} 8z\sin\theta \cdot d\theta \cdot dz = \left(\int_1^6 8z dz \right) \left(\int_0^{2\pi} \sin\theta \cdot d\theta \right)$$

$$= 4z^2 \Big|_1^6 \cdot \pi = \boxed{140\pi}$$

16. Let S be the portion of the plane $z = x$ contained in the half-cylinder of radius R depicted in Figure 17. Use Stokes' Theorem to calculate the circulation of $\mathbf{F} = (z, x, y + 2z)$ around the boundary of S (a half-ellipse) in the counterclockwise direction when viewed from above. *Hint:* Show that $\text{curl}(\mathbf{F})$ is orthogonal to the normal vector to the plane.

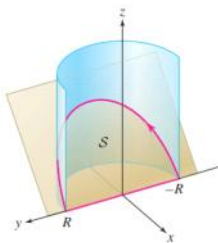


FIGURE 17

By Stokes' theorem:

$$\int_C \mathbf{F} \cdot d\mathbf{s} = \iint_S \text{curl}(\mathbf{F}) \cdot d\mathbf{S}$$

$$\text{curl}(\mathbf{F}) = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ \partial/\partial x & \partial/\partial y & \partial/\partial z \\ z & x & y + 2z \end{vmatrix} = \mathbf{i} + \mathbf{j} + \mathbf{k} = \langle 1, 1, 1 \rangle$$

$$\text{curl}(\mathbf{F}) \cdot \mathbf{n} = \langle 1, 1, 1 \rangle \cdot \langle -1, 0, 1 \rangle = -1 + 0 + 1 = 0$$

so,

$$\iint_S \text{curl}(\mathbf{F}) \cdot d\mathbf{S} = \iint_D 0 \cdot dA = 0$$

20. The magnetic field $\mathbf{B}$ due to a small current loop (which we place at the origin) is called a **magnetic dipole** (Figure 18). Let $\rho = (x^2 + y^2 + z^2)^{1/2}$. For ρ large, $\mathbf{B} = \text{curl}(\mathbf{A})$, where

$$\mathbf{A} = \left(-\frac{y}{\rho^3}, \frac{x}{\rho^3}, 0 \right)$$

(a) Let C be a horizontal circle of radius R with center $(0, 0, c)$, where c is large. Show that $\mathbf{A}$ is tangent to C .

(b) Use Stokes' Theorem to calculate the flux of $\mathbf{B}$ through C .

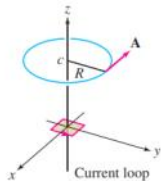


FIGURE 18

a) parameterize C by:-

$$c(t) = (R \cos t, R \sin t, c), \quad 0 \leq t \leq 2\pi$$

forget to C :-

$$c'(t) = (-R \sin t, R \cos t, 0)$$

Put in $\mathbf{A}$:-

$$\mathbf{A}(c(t)) = \left(-\frac{R \sin t}{\rho^3}, \frac{R \cos t}{\rho^3}, 0 \right)$$

b) By Stokes' theorem:-

$$\iint_S \mathbf{B} \cdot d\mathbf{S} = \iint_S \text{curl}(\mathbf{A}) \cdot d\mathbf{S} = \int_C \mathbf{A} \cdot d\mathbf{c}$$

$$\mathbf{A}(c(t)) \cdot c'(t) = |\mathbf{A}(c(t))| |c'(t)| = \sqrt{\frac{R^2 \sin^2 t}{\rho^6} + \frac{R^2 \cos^2 t}{\rho^6}} \cdot \sqrt{R^2 \sin^2 t + R^2 \cos^2 t}$$

$$= \frac{R}{\rho^3} \cdot R = \frac{R^2}{(R^2 \cos^2 t + R^2 \sin^2 t + c^2)^{3/2}}$$

$$= \frac{R^2}{(R^2 (\cos^2 t + \sin^2 t) + (c/R)^2)^{3/2}} = \frac{1}{R (1 + (c/R)^2)^{3/2}}$$

$$\iint_S \mathbf{B} \cdot d\mathbf{S} = \int_C \mathbf{A} \cdot d\mathbf{S} = \int_0^{2\pi} \frac{dt}{R (1 + (c/R)^2)^{3/2}} = \frac{2\pi}{R (1 + (c/R)^2)^{3/2}}$$

22. Let $\mathbf{F} = (-x^2 y, x, 0)$. Referring to Figure 19, let C be the closed path $ABCD$. Use Stokes' Theorem to evaluate $\int_C \mathbf{F} \cdot d\mathbf{r}$ in two ways.

First, regard C as the boundary of the rectangle with vertices A , B , C , and D . Then treat C as the boundary of the wedge-shaped box with an open top.

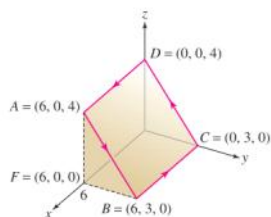


FIGURE 19

- let S_1 and S_2 be rectangles with boundary C_1 -

$$\int_{C_1} F \cdot ds = \iint_{S_1} \text{curl}(F) \cdot ds$$

and

$$\int_{C_2} F \cdot ds = \iint_{S_2} \text{curl}(F) \cdot ds$$

- Finding curl:-

$$F \times \nabla = \begin{vmatrix} i & j & k \\ \partial/\partial x & \partial/\partial y & \partial/\partial z \\ -xy & x & 0 \end{vmatrix} = \langle 0, 0, 1+x^2 \rangle$$

To find a parametrization for S_1 , we compute the equation of the plane through $A = (6, 0, 4)$, $C = (0, 3, 0)$, $D = (0, 0, 4)$.
A normal to the plane is

$$\begin{aligned} \vec{AD} \times \vec{AC} &= \langle -6, 0, 0 \rangle \times \langle -6, 3, -4 \rangle \\ &= -6i \times (-6 + 3j - 4k) = -18k - 24j = \langle 0, -24, -18 \rangle \\ &= -6 \langle 0, 4, 3 \rangle \end{aligned}$$

- We use the point-normal equation of the plane

$$\begin{aligned} 0(x-0) + 4(y-3) + 3(z-0) &= 0 \\ 4y - 12 + 3z &= 0 \\ \text{or} \\ z &= 4 - \frac{4}{3}y \end{aligned}$$

- Parametrize S_1 by:-

$$\phi(x, y) = \left(x, y, 4 - \frac{4y}{3} \right)$$

with domain $D[0, 6] \times [0, 3]$ in x - y plane.



$$\begin{aligned} \frac{\partial \phi}{\partial x} \times \frac{\partial \phi}{\partial y} &= \langle 1, 0, 0 \rangle \times \langle 0, 1, -\frac{4}{3} \rangle = i \times (j - \frac{4}{3}k) \\ &= \langle 0, \frac{4}{3}, 1 \rangle \end{aligned}$$

- upward pointing normal is:- $n = \langle 0, \frac{4}{3}, 1 \rangle$

$$\text{curl}(F) \cdot (\phi(x, y)) = \langle 0, 0, 1+x^2 \rangle$$

so,

$$\text{curl}(F) \cdot n = \langle 0, 0, 1+x^2 \rangle \cdot \langle 0, \frac{4}{3}, 1 \rangle = 1+x^2$$

- $\int_{C_1} F \cdot ds = \iint_{S_1} \text{curl}(F) \cdot ds = \iint_D (1+x^2) \cdot dA = \int_0^3 \int_0^6 (1+x^2) dx dy$

$$\begin{aligned} \int_C F \cdot ds &= \iint_{S_1} \text{curl}(F) \cdot ds = \iint_D (1+x^2) \cdot dA = \int_0^3 \int_0^6 (1+x^2) dx dy \\ &= 3 \left(x + \frac{x^3}{3} \right) \Big|_0^6 = 234 \quad \text{--- (3)} \end{aligned}$$

$$R_1: \phi_1(x, z) = (x, 0, z), \quad 0 \leq x \leq 6, \quad 0 \leq z \leq 4$$

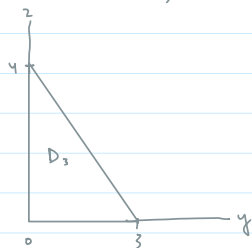
$$n_1 = \langle 0, 1, 0 \rangle$$

$$R_2: \phi_2(x, y) = (x, y, 0), \quad 0 \leq x \leq 6, \quad 0 \leq y \leq 3$$

$$n_2 = \langle 0, 0, 1 \rangle$$

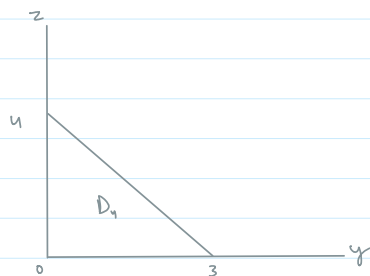
$$T_1: \phi_3(x, y) = (6, y, z), \quad (y, z) \in D_3$$

$$n_3 = \langle -1, 0, 0 \rangle$$



$$T_2: \phi_4(y, z) = (0, y, z), \quad (y, z) \in D_4$$

$$n_4 = \langle 1, 0, 0 \rangle$$



• for each surface integral-

$$\text{curl}(F) (\phi_1(x, z)) \cdot n_1 = \langle 0, 0, 1+x^2 \rangle \cdot \langle 0, 1, 0 \rangle = 0$$

$$\text{curl}(F) (\phi_2(x, y)) \cdot n_2 = \langle 0, 0, 1+x^2 \rangle \cdot \langle 0, 0, 1 \rangle = 1+x^2$$

$$\text{curl}(F) (\phi_3(x, y)) \cdot n_3 = \langle 0, 0, 1+6^2 \rangle \cdot \langle -1, 0, 0 \rangle = 0$$

$$\text{curl}(F) (\phi_4(x, y)) \cdot n_4 = \langle 0, 0, 1+6^2 \rangle \cdot \langle 1, 0, 0 \rangle = 0$$

$$\begin{aligned} \text{So, } \int_C F \cdot ds &= \iint_{S_2} \text{curl}(F) \cdot ds \\ &= \iint_{R_1} \text{curl}(F) \cdot ds + \iint_{R_2} \text{curl}(F) \cdot ds + \iint_{T_1} \text{curl}(F) \cdot ds + \iint_{T_2} \text{curl}(F) \cdot ds \end{aligned}$$

$$= \int_0^3 \int_0^6 (1+x^2) dx dy = 3 \left(x + \frac{x^3}{3} \right) \Big|_0^6 = 234 \quad - (4)$$

values in (3) & (4) match.

24. Let $\mathbf{F} = \langle -z^2, 2zx, 4y - x^2 \rangle$, and let C be a simple closed curve in the plane $x + y + z = 4$ that encloses a region of area 16 (Figure 20).

Calculate $\oint_C \mathbf{F} \cdot d\mathbf{r}$, where C is oriented in the counterclockwise direction (when viewed from above the plane).

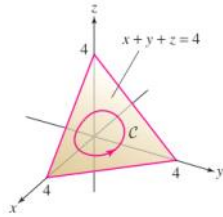


FIGURE 20

• find curl:-

$$\text{Curl}(\mathbf{F}) = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ \partial/\partial x & \partial/\partial y & \partial/\partial z \\ -z^2 & 2zx & 4y - x^2 \end{vmatrix} = \langle 4 - 2x, 2x - 2z, 2z \rangle$$

• Plane $x + y + z = 4$ has parametrization

$$\phi(x, y) = \langle x, y, 4 - x - y \rangle$$

- normal will be coefficient = $\langle 1, 1, 1 \rangle$

$$\begin{aligned} \text{Curl}(\mathbf{F}) \cdot (\phi(x, y)) &= \langle 4 - 2x, 2x - 2(4 - x - y), 2(4 - x - y) \rangle \\ &= \langle 4 - 2x, -8 + 4x + 2y, 8 - 2x - 2y \rangle \end{aligned}$$

$$\text{Curl}(\mathbf{F}) \cdot \mathbf{n} = 4 - 2x - 8 + 4x + 2y + 8 - 2x - 2y = 4$$

So,

$$\begin{aligned} \int_C \mathbf{F} \cdot d\mathbf{s} &= \iint_D \text{Curl}(\mathbf{F}) \cdot \mathbf{n} \, dA = \iint_D 4 \, dA = 4 \cdot \text{Area}(D) \\ &= 4 \cdot \frac{16}{2} = \boxed{32} \text{ units} \end{aligned}$$

25. Let $\mathbf{F} = \langle y^2, x^2, z^2 \rangle$. Show that

$$\int_{C_1} \mathbf{F} \cdot d\mathbf{r} = \int_{C_2} \mathbf{F} \cdot d\mathbf{r}$$

for any two closed curves lying on a cylinder whose central axis is the z -axis (Figure 21).

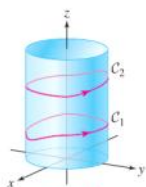


FIGURE 21

- As C_1 & C_2 have same orientation-

$$\int_{C_1} \mathbf{F} \cdot d\mathbf{s} - \int_{C_2} \mathbf{F} \cdot d\mathbf{s} = \iint_S \text{curl}(\mathbf{F}) \cdot d\mathbf{s} \quad \text{--- (i)}$$

$$\text{curl}(\mathbf{F}) = \left\langle \frac{\partial F_3}{\partial y} - \frac{\partial F_2}{\partial z}, \frac{\partial F_1}{\partial z} - \frac{\partial F_3}{\partial x}, \frac{\partial F_2}{\partial x} - \frac{\partial F_1}{\partial y} \right\rangle = \langle 0, 0, 2x - 2y \rangle$$

$$\phi(\theta, z) = \langle R \cos \theta, R \sin \theta, z \rangle$$

$$\mathbf{n} = \langle R \cos \theta, R \sin \theta, 0 \rangle$$

$$\text{curl}(\mathbf{F}) \cdot \mathbf{n} = 2R \langle 0, 0, \cos \theta - \sin \theta \rangle \cdot R \langle \cos \theta, \sin \theta, 0 \rangle = 2R^2 (0 + 0 + 0) = 0$$

comparing with eq (i):-

$$\int_{C_1} \mathbf{F} \cdot d\mathbf{s} - \int_{C_2} \mathbf{F} \cdot d\mathbf{s} = \iint_S \text{curl}(\mathbf{F}) \cdot d\mathbf{s} = \iint_D 0 \cdot d\theta \cdot dz = 0$$

So, $\boxed{\int_{C_1} \mathbf{F} \cdot d\mathbf{s} = \int_{C_2} \mathbf{F} \cdot d\mathbf{s}}$ Hence proved.

31. Prove the following Product Rule:

$$\text{curl}(f\mathbf{F}) = f \text{curl}(\mathbf{F}) + \nabla f \times \mathbf{F}$$

$$\text{curl}(f\mathbf{F}) = f \text{curl}(\mathbf{F}) + \nabla f \times \mathbf{F}$$

- we evaluate the curl of $f\mathbf{F}$. since $f\mathbf{F} = \langle fF_1, fF_2, fF_3 \rangle$
using product rule-

$$\text{curl}(f\mathbf{F}) = \left\langle \frac{\partial}{\partial y}(fF_3) - \frac{\partial}{\partial z}(fF_2), \frac{\partial}{\partial z}(fF_1) - \frac{\partial}{\partial x}(fF_3), \frac{\partial}{\partial x}(fF_2) - \frac{\partial}{\partial y}(fF_1) \right\rangle$$

$$= f \left\langle \frac{\partial F_3}{\partial y} - \frac{\partial F_2}{\partial z}, \frac{\partial F_1}{\partial z} - \frac{\partial F_3}{\partial x}, \frac{\partial F_2}{\partial x} - \frac{\partial F_1}{\partial y} \right\rangle + \left\langle \frac{\partial f}{\partial y} F_3 - \frac{\partial f}{\partial z} F_2, \frac{\partial f}{\partial z} F_1 - \frac{\partial f}{\partial x} F_3, \frac{\partial f}{\partial x} F_2 - \frac{\partial f}{\partial y} F_1 \right\rangle \quad \text{--- (i)}$$

- The vector in the first term is $\text{curl}(\mathbf{F})$.

- The vector in the first term is $\text{curl}(F)$.

$$\nabla f \times F = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ \frac{\partial f}{\partial x} & \frac{\partial f}{\partial y} & \frac{\partial f}{\partial z} \\ F_1 & F_2 & F_3 \end{vmatrix} = \left\langle \frac{\partial f}{\partial y} F_3 - \frac{\partial f}{\partial z} F_2, \frac{\partial f}{\partial z} F_1 - \frac{\partial f}{\partial x} F_3, \frac{\partial f}{\partial x} F_2 - \frac{\partial f}{\partial y} F_1 \right\rangle$$

Then we have :-

$$(2) \rightarrow \text{curl}(fF) = f \text{curl}(F) + \nabla f \times F$$

SECTION 17.3

7. $F(x, y, z) = (xy^2, yz^2, zx^2)$, S is the boundary of the cylinder given by $x^2 + y^2 \leq 4$, $0 \leq z \leq 3$.

- Calculate divergence:-

$$\text{div}(F) = \frac{\partial}{\partial x}(xy^2) + \frac{\partial}{\partial y}(yz^2) + \frac{\partial}{\partial z}(zx^2)$$

So,

$$\text{div}(F) = y^2 + z^2 + x^2$$

- parameterize:-

$$\begin{aligned} x &= r \cos \theta & 0 \leq r \leq 2 \\ y &= r \sin \theta & 0 \leq \theta \leq 2\pi \\ z &= z & 0 \leq z \leq 3 \\ dV &= r \, dr \, d\theta \, dz \end{aligned}$$

- In cylindrical coordinates $\text{div}(F) = r^2 + z^2$

$$\iiint_E (x^2 + y^2 + z^2) \, dV = \int_0^{2\pi} \int_0^2 \int_0^3 (r^2 + z^2) \cdot r \, dz \, dr \, d\theta$$

after calculation we get.

$$\iint_S F \cdot d\mathbf{s} = \boxed{60\pi} \text{ Ans.}$$

9. $F(x, y, z) = (x + z^2, xz + y^2, zx - y)$, S is the surface that bounds the solid region with boundary given by the parabolic cylinder $z = 1 - x^2$, and the planes $z = 0$, $y = 0$ and $z + y = 5$.

- Calculate Div:-

$$\text{div}(F) = \frac{\partial}{\partial x}(x + z^2) + \frac{\partial}{\partial y}(xz + y^2) + \frac{\partial}{\partial z}(zx - y)$$

$$\text{div}(F) = 1 + 2y + x$$

So,

$$-1 \leq x \leq 1$$

$$0 \leq y \leq 5-z$$

$$0 \leq z \leq 1-x^2$$

- Integrate wrt of respect to y:-

$$\begin{aligned} \int_0^{(5-z)} (1+2y+x) \cdot dy &= (5-z) [1 + (5-z) + x] \\ &= (5-z)(6-z+x) \end{aligned}$$

- Integrate wrt z:-

$$\begin{aligned} \int_0^{(1-x^2)} (5-z)(6-z+x) \cdot dz \\ &= \int_0^{(1-x^2)} [30 + 5x - 11z - xz + z^2] \cdot dz \\ &= \left[30z + 5xz - 11z^2/2 - xz^2/2 + z^3/3 \right]_0^{(1-x^2)} \\ &= 30(1-x^2) + 5x(1-x^2) - \left(\frac{11}{2}\right)(1-x^2)^2 - \left(\frac{x}{2}\right)(1-x^2)^2 + \frac{1}{3}(x^3)^3 \end{aligned}$$

- Integrate wrt x:-

$$= \int_{-1}^1 \left[30(1-x^2) + 5x(1-x^2) - \left(\frac{11}{2}\right)(1-x^2)^2 - \left(\frac{x}{2}\right)(1-x^2)^2 + \left(\frac{1}{3}\right)(1-x^2)^3 \right] dx$$

after calculation we get.

$$\iint_S F \cdot ds = 40 + 88/15 + 32/105 = \boxed{41.71}$$

15. $F(x, y, z) = (x + y, z, z - x)$, S is the boundary of the region between the paraboloid $z = 9 - x^2 - y^2$ and the xy -plane.

$$\text{div}(F) = \frac{\partial}{\partial x}(x+y) + \frac{\partial}{\partial y}(z) + \frac{\partial}{\partial z}(z-x) = 1+0+1=2$$

$$\begin{aligned} \text{So, } \iint_S F \cdot ds &= \iiint_W 2 \cdot dV = \iiint_0^{9-x^2-y^2} 2 \cdot dz \cdot dx \cdot dy \\ &= \iint_W 2(9-x^2-y^2) \cdot dx \cdot dy \end{aligned}$$

- Convert to polar coordinates:-

$$x = r \cos \theta, \quad y = r \sin \theta, \quad 0 \leq r \leq 3 \quad 0 \leq \theta \leq 2\pi$$

$$x = r \cos \theta, \quad y = r \sin \theta, \quad 0 \leq r \leq 3 \quad 0 \leq \theta \leq 2\pi$$

$$\iint_S F \cdot d\mathbf{S} = \int_0^{2\pi} \int_0^3 2(9-r^2) r \cdot dr \cdot d\theta$$

$$= 4\pi \left(\frac{9r^2}{2} - \frac{r^4}{4} \Big|_0^3 \right) = 81\pi$$

17. Calculate the flux of the vector field $\mathbf{F} = 2xy\mathbf{i} - y^2\mathbf{j} + \mathbf{k}$ through the surface S in Figure 18. *Hint:* Apply the Divergence Theorem to the closed surface consisting of S and the unit disk.

$$\operatorname{div}(\mathbf{F}) = \frac{\partial}{\partial x}(2xy) + \frac{\partial}{\partial y}(-y^2) + \frac{\partial}{\partial z}(1) = 2y - 2y + 0 = 0$$

$$\therefore \iint_S \mathbf{F} \cdot d\mathbf{S} = - \iint_D \mathbf{F} \cdot d\mathbf{S}$$

$$\phi(r, \theta) = (r \cos \theta, r \sin \theta, 0)$$

$$\mathbf{F}(\phi(r, \theta)) = \langle 2r^2 \cos \theta \sin \theta, -r^2 \sin^2 \theta, 1 \rangle$$

$$\phi_r(r, \theta) = \langle \cos \theta, \sin \theta, 0 \rangle \quad \phi_\theta(r, \theta) = \langle -r \sin \theta, r \cos \theta, 0 \rangle$$

and

$$\phi_r \times \phi_\theta = \langle 0, 0, r \rangle$$

$$\mathbf{F} \cdot d\mathbf{S} = \mathbf{F}(\phi(r, \theta)) \cdot (\phi_r \times \phi_\theta) \cdot dr \cdot d\theta = r \cdot dr \cdot d\theta$$

$$= \int_0^{2\pi} \int_0^1 r \cdot dr \cdot d\theta = \boxed{\pi} \text{ Am.}$$

18. Let S_1 be the closed surface consisting of S in Figure 18 together with the unit disk. Find the volume enclosed by S_1 , assuming that

$$\iint_{S_1} (x, 2y, 3z) \cdot d\mathbf{S} = 72$$

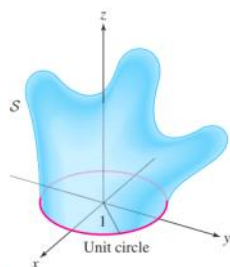


FIGURE 18 Surface S whose boundary is the unit circle.

$$\operatorname{div}(\mathbf{F}) = \frac{\partial}{\partial x}(x) + \frac{\partial}{\partial y}(2y) + \frac{\partial}{\partial z}(3z) = 1 + 2 + 3 = 6$$

$$\text{div}(F) = \frac{\partial}{\partial x}(x) + \frac{\partial}{\partial y}(2y) + \frac{\partial}{\partial z}(3z) = 1 + 2 + 3 = 6$$

$$\iint_S (x, 2y, 3z) \cdot d\mathbf{s} = \iiint_W \text{div } F \, dV = \iiint_W 6 \, dV = 6 \cdot \text{volume}(W)$$

So,

$$\boxed{\text{volume}(W) = 12} \text{ Ans.}$$

23. Let W be the region in Figure 19 bounded by the cylinder $x^2 + y^2 = 4$, the plane $z = x + 1$, and the xy -plane. Use the Divergence Theorem to compute the flux of $\mathbf{F}(x, y, z) = (z, x, y + z^2)$ through the boundary of W .

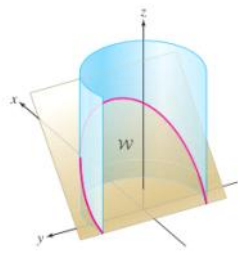


FIGURE 19

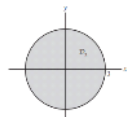
$$\text{div}(F) = \frac{\partial}{\partial z}(z) + \frac{\partial}{\partial y}(x) + \frac{\partial}{\partial x}(y + z^2) = 2z$$

$$\iint_S F \cdot d\mathbf{s} = \iiint_W \text{div}(F) \, dV = \iiint_W 2z \, dV$$

So,

$$\iint_S F \cdot d\mathbf{s} = \iiint_W 2z \, dV = \iint_D \int_0^{x+1} 2z \, dz \, dx \, dy$$

$$= \iint_D 2z \Big|_{z=0}^{x+1} \, dx \, dy = \iint_D (x+1)^2 \, dx \, dy$$



$$\iint_S F \cdot d\mathbf{s} = \iint_{D_2} (x+1)^2 \, dx \, dy - \iint_{D_1} (x+1)^2 \, dx \, dy$$

• Integral for D_1 :-

$$\therefore \iint_{D_1} (x+1)^2 \, dx \, dy = \int_0^\pi \int_0^2 (r \cos \theta + 1)^2 r \, dr \, d\theta$$

after solving we get " 8π "

• Computing integral for D_2 :-

$$\frac{2\pi}{3} \leq \theta \leq \pi, \quad -1 \leq r \leq 2$$

So

$$\iint (x+1)^2 \, dx \, dy = \int_{2\pi/3}^\pi \int_{-1}^2 (r \cos \theta + 1)^2 r \, dr \, d\theta$$

So

$$\iint_{D_1} (x+1)^2 \, dx \, dy = 2 \int_{\pi/3}^{\pi} \int_{-1/\cos\theta}^2 (r\cos\theta+1)^2 r \, dr \, d\theta$$

$$= \int_{\pi/3}^{\pi} \int_{-1/\cos\theta}^2 (r^3 \cos^3\theta + 2r^2 \cos\theta + r) \, dr \, d\theta$$

$$= 2 \int_{\pi/3}^{\pi} \left[\frac{r^4}{4} \cos^3\theta + \frac{2r^3}{3} \cos\theta + \frac{r^2}{2} \right]_{-1/\cos\theta}^2 d\theta$$

$$= 8\pi - 2 \left(-\frac{\sqrt{3}}{2} + \frac{8\pi}{3} + \frac{16\sqrt{3}}{6} + \frac{\sqrt{3}}{12} \right)$$

$$= \frac{8\pi}{3} - \frac{9\sqrt{3}}{2}$$

So we have, finally-

$$\iint_S \mathbf{F} \cdot d\mathbf{s} = 8\pi - \left(\frac{8\pi}{3} - \frac{9\sqrt{3}}{2} \right) = \boxed{24.550}$$

34. Prove the identity

$$\text{curl}(\text{curl}(\mathbf{F})) = \nabla(\text{div}(\mathbf{F})) - \Delta \mathbf{F}$$

where $\Delta \mathbf{F}$ denotes $(\Delta F_1, \Delta F_2, \Delta F_3)$.

$$\text{curl}(\mathbf{F}) = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ F_1 & F_2 & F_3 \end{vmatrix}$$

$$= \left\langle \frac{\partial F_3}{\partial y} - \frac{\partial F_2}{\partial z}, \frac{\partial F_1}{\partial z} - \frac{\partial F_3}{\partial x}, \frac{\partial F_2}{\partial x} - \frac{\partial F_1}{\partial y} \right\rangle$$

So,

$$\text{curl}(\text{curl}(\mathbf{F})) = \left\langle \frac{\partial}{\partial y} \left(\frac{\partial F_2}{\partial x} - \frac{\partial F_1}{\partial z} \right) - \frac{\partial}{\partial z} \left(\frac{\partial F_1}{\partial z} - \frac{\partial F_3}{\partial x} \right), \frac{\partial}{\partial z} \left(\frac{\partial F_3}{\partial y} - \frac{\partial F_2}{\partial z} \right) - \frac{\partial}{\partial x} \left(\frac{\partial F_2}{\partial x} - \frac{\partial F_1}{\partial y} \right), \right.$$

$$\left. \frac{\partial}{\partial x} \left(\frac{\partial F_1}{\partial z} - \frac{\partial F_3}{\partial x} \right) - \frac{\partial}{\partial y} \left(\frac{\partial F_3}{\partial y} - \frac{\partial F_2}{\partial z} \right) \right\rangle$$

$$= \left\langle \frac{\partial^2 F_2}{\partial x \partial y} + \frac{\partial^2 F_3}{\partial x \partial z} - \frac{\partial^2 F_1}{\partial y^2} - \frac{\partial^2 F_1}{\partial z^2}, \frac{\partial^2 F_1}{\partial y \partial x} + \frac{\partial^2 F_3}{\partial y \partial z} + \frac{\partial^2 F_2}{\partial x^2} - \frac{\partial^2 F_2}{\partial z^2}, \right.$$

$$\left. \frac{\partial^2 F_1}{\partial z \partial x} + \frac{\partial^2 F_2}{\partial z \partial y} - \frac{\partial^2 F_3}{\partial x^2} - \frac{\partial^2 F_3}{\partial y^2} \right\rangle$$

Now, compute R.H.S:-

$$\nabla(\text{div}(\mathbf{F})) = \nabla \left(\frac{\partial F_1}{\partial x} + \frac{\partial F_2}{\partial y} + \frac{\partial F_3}{\partial z} \right)$$

after simplifying-

$$\nabla(\text{div}(\mathbf{F})) = \nabla \left(\frac{\partial F_1}{\partial x} + \frac{\partial F_2}{\partial y} + \frac{\partial F_3}{\partial z} \right)$$

after solving:-

$$= \left\langle \frac{\partial^2 F_1}{\partial x^2} + \frac{\partial^2 F_2}{\partial x \partial y} + \frac{\partial^2 F_3}{\partial x \partial z}, \frac{\partial^2 F_1}{\partial y \partial x} + \frac{\partial^2 F_2}{\partial y^2} + \frac{\partial^2 F_3}{\partial y \partial z}, \frac{\partial^2 F_1}{\partial z \partial x} + \frac{\partial^2 F_2}{\partial z \partial y} + \frac{\partial^2 F_3}{\partial z^2} \right\rangle$$

Therefore-

$$\nabla(\text{div}(\mathbf{F})) - \Delta \mathbf{F} =$$

$$= \left\langle \frac{\partial^2 F_2}{\partial x \partial y} + \frac{\partial^2 F_3}{\partial x \partial z} - \frac{\partial^2 F_1}{\partial y^2} - \frac{\partial^2 F_1}{\partial z^2}, \frac{\partial^2 F_1}{\partial y \partial x} + \frac{\partial^2 F_3}{\partial y \partial z} + \frac{\partial^2 F_2}{\partial x^2} - \frac{\partial^2 F_2}{\partial z^2}, \right.$$

$$\left. \frac{\partial^2 F_1}{\partial z \partial x} + \frac{\partial^2 F_2}{\partial z \partial y} - \frac{\partial^2 F_3}{\partial x^2} - \frac{\partial^2 F_3}{\partial y^2} \right\rangle$$

as the signs are equal, hence proved.

36. Let $\mathbf{F} = r^n \mathbf{e}_r$, where n is any number, $r = (x^2 + y^2 + z^2)^{1/2}$, and $\mathbf{e}_r = r^{-1} \langle x, y, z \rangle$ is the unit radial vector.

(a) Calculate $\text{div}(\mathbf{F})$.

(b) Calculate the flux of $\mathbf{F}$ through the surface of a sphere of radius R centered at the origin. For which values of n is this flux independent of R ?

(c) Prove that $\nabla(r^n) = n r^{n-1} \mathbf{e}_r$.

(d) Use (c) to show that $\mathbf{F}$ is conservative for $n \neq -1$. Then show that $\mathbf{F} = r^{-1} \mathbf{e}_r$ is also conservative by computing the gradient of $\ln r$.

(e) What is the value of $\int_C \mathbf{F} \cdot d\mathbf{s}$, where C is a closed curve that does not pass through the origin?

(f) Find the values of n for which the function $\phi = r^n$ is harmonic.

$$a) F(x, y, z) = r^n r^{-1} \langle x, y, z \rangle = (x^2 + y^2 + z^2)^{(n-1)/2} (x + y + z)$$

$$\frac{\partial F_1}{\partial x} = \frac{\partial}{\partial x} \left((x^2 + y^2 + z^2)^{(n-1)/2} x \right) = \left(\frac{n-1}{2} \right) (x^2 + y^2 + z^2)^{(n-3)/2} \cdot 2x \cdot x + (x^2 + y^2 + z^2)^{(n-1)/2}$$

$$= (x^2 + y^2 + z^2)^{(n-3)/2} \left((n-1)x^2 + x^2 + y^2 + z^2 \right) = (x^2 + y^2 + z^2)^{(n-3)/2} (nx^2 + y^2 + z^2)$$

Similarly-

$$\frac{\partial F_2}{\partial y} = (x^2 + y^2 + z^2)^{(n-3)/2} (x^2 + ny^2 + z^2)$$

and

$$\frac{\partial F_3}{\partial z} = (x^2 + y^2 + z^2)^{(n-3)/2} (x^2 + y^2 + nz^2)$$

• $\text{div}(\mathbf{F})$ is the sum, which will be equal to :-

$$\text{div}(\mathbf{F}) = \frac{\partial F_1}{\partial x} + \frac{\partial F_2}{\partial y} + \frac{\partial F_3}{\partial z} = (n+2) r^{n-1}$$

$$b) \iint_S F \cdot d\mathbf{s} = \iiint_W \operatorname{div}(F) \cdot dV = \iiint_W (n+2)r^{n-1} \cdot dV$$

$$\iint_S F \cdot d\mathbf{s} = \int_0^{2\pi} \int_0^\pi \int_0^R (n+2)r^{n-1} r^2 \sin\phi \, dr \, d\phi \, d\theta$$

$$= 2\pi (n+2) \int_0^\pi \int_0^R r^{n+1} \sin\phi \, dr \, d\phi$$

after solving:-

$$= 4\pi (n+2) \frac{R^{n+2}}{n+2} = 4\pi R^{n+2}$$

• Now consider $n < 3$, we evaluate the surface integral directly using parametrization. $\phi(\theta, \phi) = (R \cos\theta \sin\phi, R \sin\theta \sin\phi, R \cos\phi)$
 $0 \leq \theta \leq 2\pi \quad 0 \leq \phi \leq \pi$

$$\mathbf{n} = R^2 \sin\phi \, \mathbf{e}_r$$

$$F(\phi(\theta, \phi)) \cdot \mathbf{n} = r^n \mathbf{e}_r \cdot R^2 \sin\phi \mathbf{e}_r = r^n R^2 \sin\phi$$

$$= R^2 \sin\phi (R^1 \cos^2\theta \sin^2\phi + R^1 \sin^2\theta \sin^2\phi + R^1 \cos^2\phi)^{n/2}$$

$$= R^{n+2} \sin\phi$$

$$\text{Hence :- } \iint_S F \cdot d\mathbf{s} = \int_0^{2\pi} \int_0^\pi R^{n+2} \sin\phi \, d\phi \, d\theta = \boxed{4\pi R^{n+2}}$$

Ans

$$c) \frac{\partial r}{\partial x} = \frac{\partial}{\partial x} (x^2 + y^2 + z^2)^{1/2} = \frac{1}{2} (x^2 + y^2 + z^2)^{-1/2} \cdot 2x = \frac{x}{r}$$

$$\frac{\partial r}{\partial y} = \frac{y}{r}, \quad \frac{\partial r}{\partial z} = \frac{z}{r}$$

• Thus,

$$\nabla(r^n) = \left(n r^{n-2} x, n r^{n-2} y, n r^{n-2} z \right) = n r^{n-1} \mathbf{e}_r$$

d) Replacing n by $n+1$ in the equality of part (c)

$$\nabla(r^{n+1}) = (n+1)r^n \mathbf{e}_r = (n+1)F$$

hence

if $n \neq -1$, then

$$F = \nabla \left(\frac{r^{n+1}}{n+1} \right)$$

Now show that $F = r^{-1} \mathbf{e}_r$ is also conservative:-

$$\frac{\partial}{\partial x} (r^{-1}) = \frac{1}{r} \frac{\partial r}{\partial x} = \frac{1}{r} \frac{x}{r} = \frac{x}{r^2}$$

$$\frac{\partial}{\partial x} (u r) = \frac{1}{r} \frac{\partial r}{\partial x} = \frac{1}{r} \frac{x}{r} = \frac{x}{r^2}$$

$$\frac{\partial}{\partial y} (u r) = \frac{y}{r^2}, \quad \frac{\partial}{\partial z} (u r) = \frac{z}{r^2}$$

here for

$$\nabla(u r) = r^{-1} \mathbf{e}_r = 'F'$$

e)

Since F is conservative, the line integral of F is zero over every closed curve in the domain of F , that is, over every closed curve not passing through the origin

f)

$$\operatorname{div}(\nabla(r^n)) = 0$$

$$\operatorname{div}(n r^{n-1} \mathbf{e}_r) = n \operatorname{div}(r^{n-1} \mathbf{e}_r) = 0$$

$$\text{either } \operatorname{div}(r^{n-1} \mathbf{e}_r) = 0 \quad \text{or} \quad n = 0$$

• we put (a) for (n):

$$\operatorname{div}(r^{n-1} \mathbf{e}_r) = (n+1) r^{n-2} = 0 \quad \begin{array}{l} (n=-1) \\ (n=0) \end{array}$$

• we conclude that $\phi = r^n$ is harmonic for $n=-1$
or $n=0$